

AIRPORT SYSTEM

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ABSTRACT

- The main theme of our project will be around the departure area of an airport. It is common for passengers to lose time at the airport in checking area. Some may even miss the flight in which they bought. Also, weight luggage's and bags are another time-consuming consideration. To please all passengers, it is important to minimize airport waiting times and maximize the efficiency of an airport staff.



COMPONENTS OF THE SYSTEM

- **System:** Airport
- **Entities:** Passengers, Airport personnel
- **Attributes for passengers:** Arrival time for domestic terminal, Arrival time for international terminal, Required service for domestic terminal, Required service for international terminal
- **Attributes for personnel:** Service time for domestic terminal employee, Service time for international terminal employee



COMPONENTS OF THE SYSTEM

- **Activities:** Service , service time(domestic and international terminal employee), interarrival time (pasennger in domestic and international terminal)
- **Events:** Starting the service, Ending the service, Arrival, Departure, domestic and international terminal
- **State Variables:** Status of personnel (busy or idle), Status of passenger (wait or not)
- **Output variables:** Number of passengers who wait in the system.



OBJECTIVE OF THE SYSTEM

- Maximize customer satisfaction by minimizing the waiting time and the number of people waiting in the system. These conditions are provided by optimizing the number of reservations to be made and allowing a certain number of people to get in airport during the day.



DATA COLLECTION

- **Stochastic Datas:**

- Interarrival times for the passengers both domestic and international terminal .
- Service time for employee both domestic and international terminal.

- **Deterministic Data:**

- Number of airport personel.



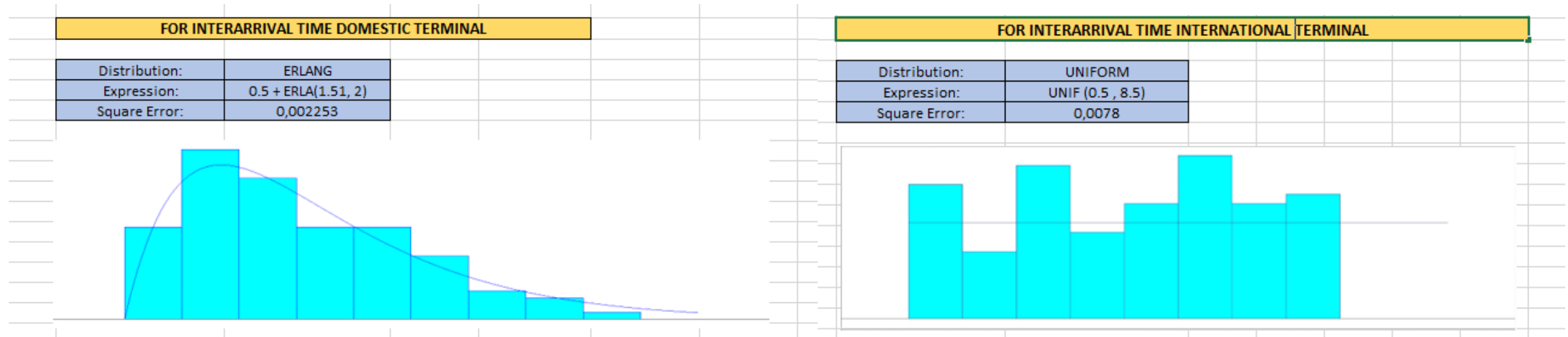
DATA COLLECTION

- Since real data could not be reached in this model made for airport ticket or baggage queue, random values were studied. Input analyzer was used to calculate the time interarrivals of the waiting passenger and the service times of the airport personnel both domestic and international terminal.



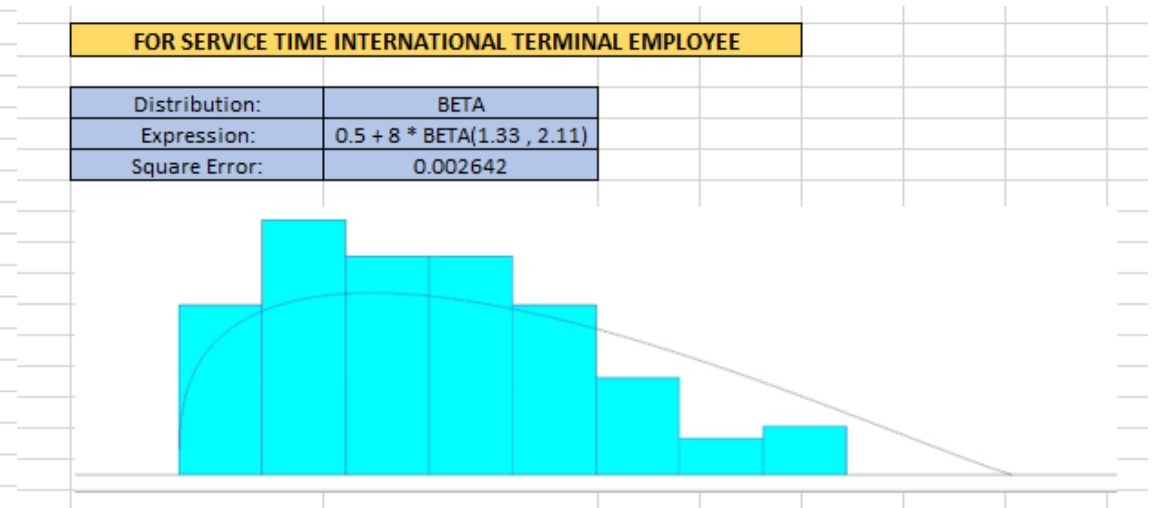
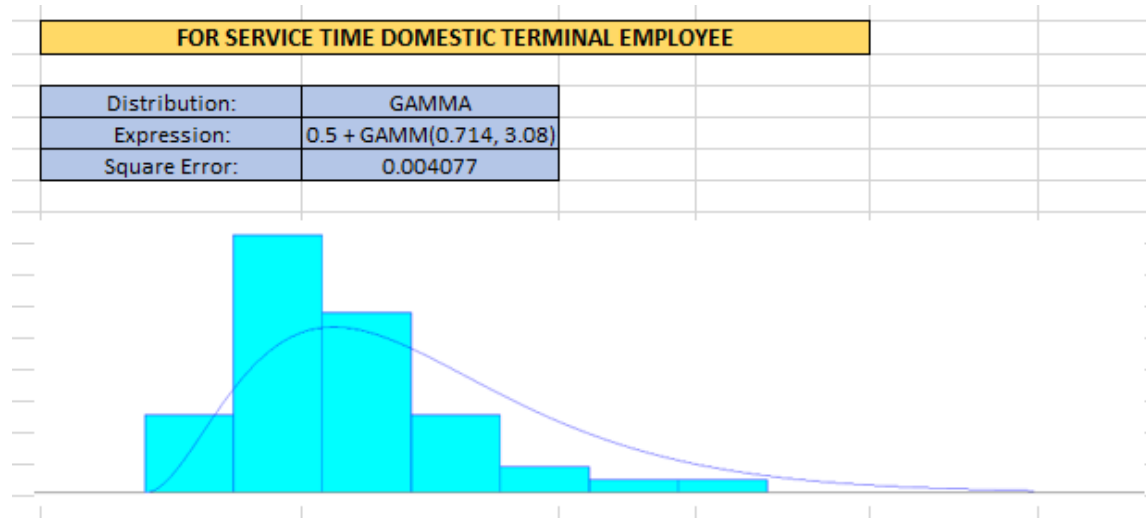
DISTRIBUTION DETERMINATION USING INPUT ANALYZER

- First of all, distribution was determined in the input analyzer for passenger interarrival time for domestic and international terminal.

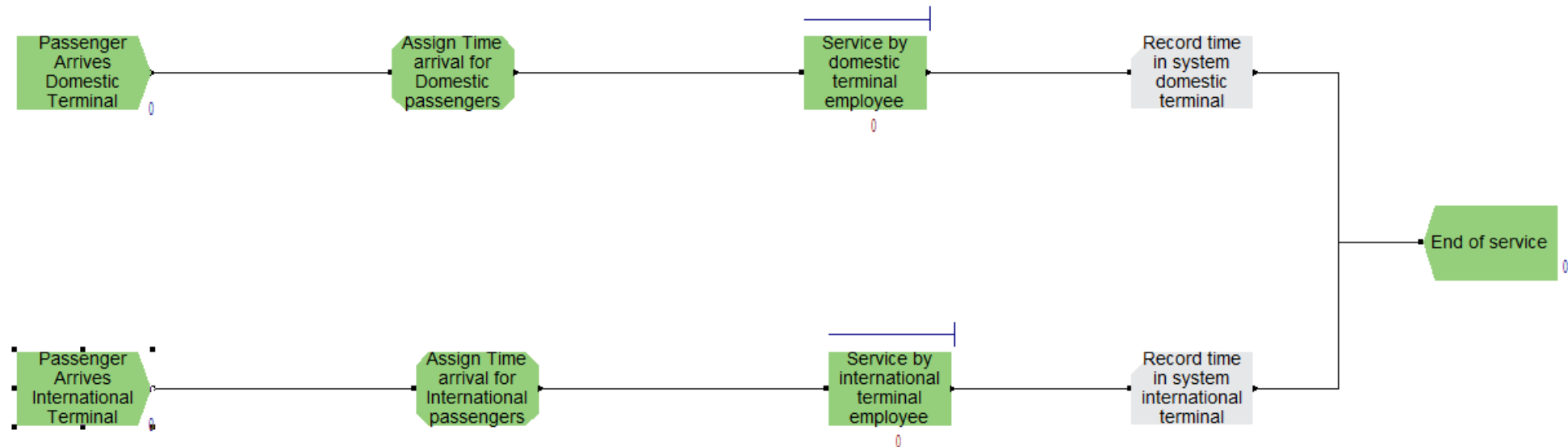


DISTRIBUTION DETERMINATION USING INPUT ANALYZER

- The service time distributions of the airport employees were determined for both international and domestic terminal.



ARENA MODEL



ARENA MODEL

- Two separate create modules for the passenger arrivals domestic terminal and the passenger arrivals international terminal were created with the distribution values found by input analysis.
- Separate assign modules are assigned to these two separate create modules.
- Using the distributions of the service times determined by the input analysis, two separate process modules for the service by domestic terminal employee and the service by international terminal employee were connected to the assign.
- The times of the events passed in the system were kept with the record models connected to the process modules.
- Finally, these separate entities leave the system with the common dispos model.



ARENA RESULTS

- Category Overview

12:43:46	Category Overview	Aralık 25, 2021
Unnamed Project		
Replications: 1	Time Units: Hours	
Key Performance Indicators		
System	Average	
Number Out	241	



ARENA RESULTS

■ Category by Replication

VA time

Domestic passengers: **2.8 minutes**

International passengers: **3.7 minutes**

Wait time

Domestic passengers: **2.1 minutes**

International passengers: **2.4 minutes**

Total time

Domestic passengers: **4.9 minutes**

International passengers: **6.1 minutes**

14:06:57

Category by Replication

Aralık 25, 2021

Unnamed Project

Replications: 1

Replication 1

Start Time: 0,00 Stop Time: 8,00 Time Units: Hours

Entity

Time

VA Time	Average	Half Width	Minimum	Maximum
Domestic Passengers	0.04651790	(Insufficient)	0.01504550	0.1029
International passengers	0.06177459	(Insufficient)	0.00863942	0.1356
NVA Time	Average	Half Width	Minimum	Maximum
Domestic Passengers	0	(Insufficient)	0	0
International passengers	0	(Insufficient)	0	0
Wait Time	Average	Half Width	Minimum	Maximum
Domestic Passengers	0.03440592	(Insufficient)	0	0.1271
International passengers	0.04007252	(Insufficient)	0	0.1743
Transfer Time	Average	Half Width	Minimum	Maximum
Domestic Passengers	0	(Insufficient)	0	0
International passengers	0	(Insufficient)	0	0
Other Time	Average	Half Width	Minimum	Maximum
Domestic Passengers	0	(Insufficient)	0	0
International passengers	0	(Insufficient)	0	0
Total Time	Average	Half Width	Minimum	Maximum
Domestic Passengers	0.08092382	(Insufficient)	0.01504550	0.1925
International passengers	0.1018	(Insufficient)	0.01696297	0.2124

Other

Number In	Value
Domestic Passengers	136
International passengers	109
Number Out	Value
Domestic Passengers	134
International passengers	107

Model Filename: C:\Users\mnmeal\Desktop\projemodel\III

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ARENA RESULTS

■ Queues

12:45:15

Queues

Aralık 25, 2021

Unnamed Project

Replications: 1

Replication 1

Start Time: 0,00

Stop Time: 8,00

Time Units: Hours

Queue Detail Summary

Time

	<u>Waiting Time</u>
Service by domestic terminal employee.Queue	0.03
Service by international terminal employee.Queue	0.04

Other

	<u>Number Waiting</u>
Service by airport employee 2.Queue	0.00
Service by airport employee.Queue	0.00
Service by domestic terminal employee.Queue	0.59
Service by international terminal employee.Queue	0.54



ARENA RESULTS

■ User Specified

Record time in system domestic terminal average: **4.9 minutes**

Record time in system international terminal average: **6.1 minutes**

14:02:23	User Specified			Aralık 26, 2021
Unnamed Project			Replications: 1	
Replication 1		Start Time: 0,00	Stop Time: 8,00	Time Units: Hours
Tally				
Interval	Average	Half Width	Minimum	Maximum
Record time in system domestic terminal	0.08092382	(Insufficient)	0.01504550	0.1925
Record time in system international terminal	0.1018	(Insufficient)	0.01696297	0.2124
Counter				
Count	Value			
number of domestic passengers	134.00			
Number of international passengers	107.00			



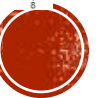
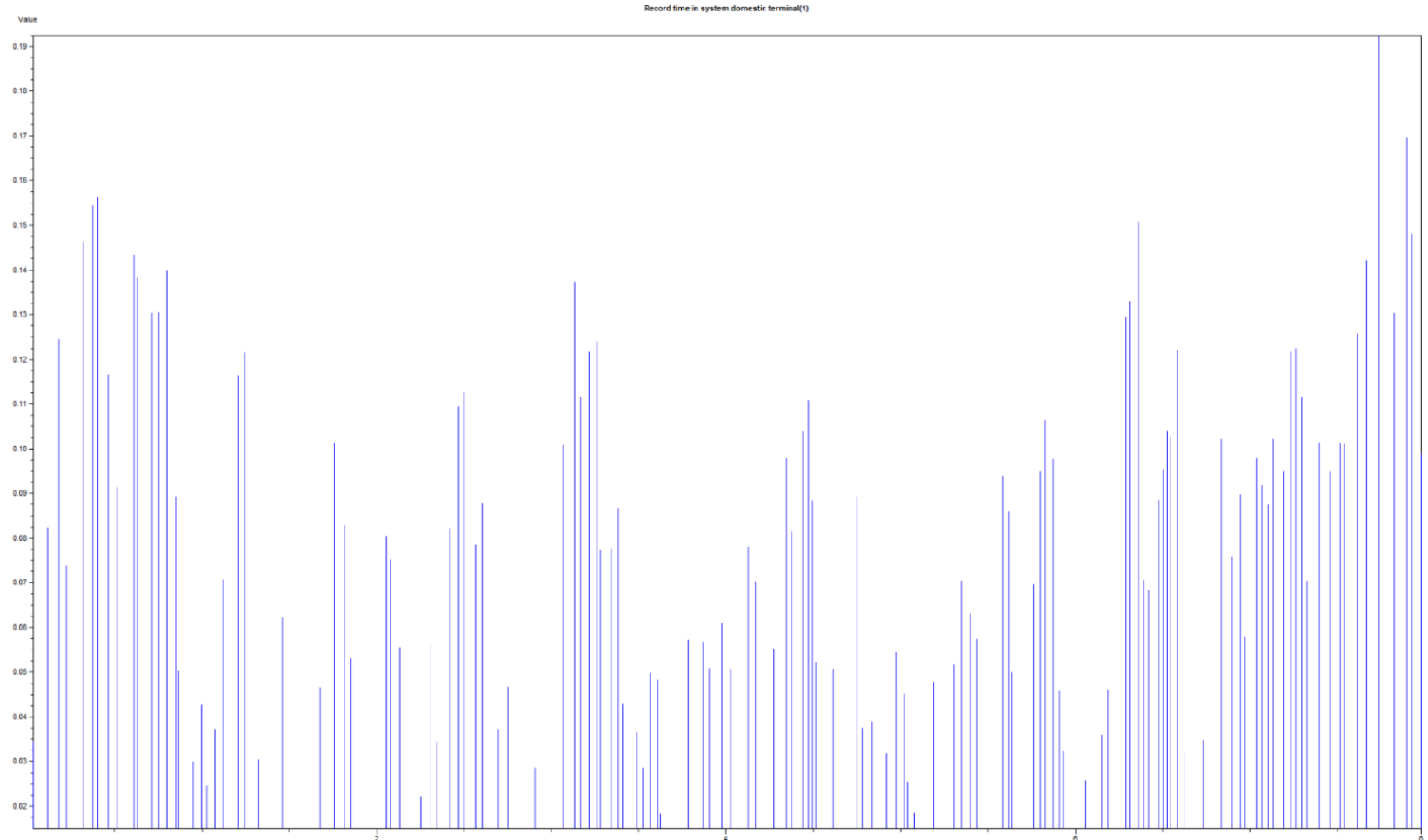
OUTPUT ANALYSIS

- Output analysis was used for the analysis of the collected arena results and gives statistical information.



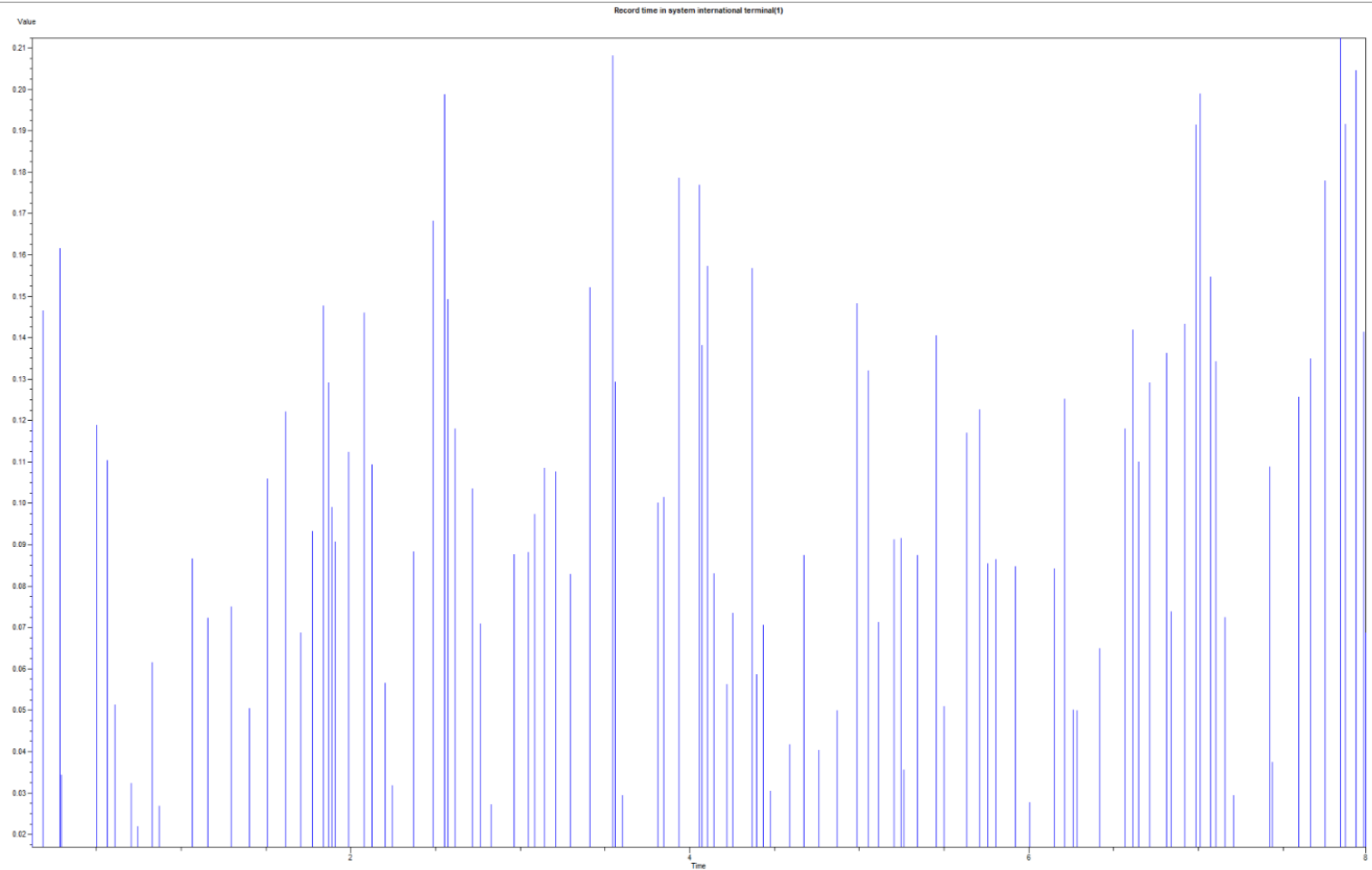
OUTPUT ANALYSIS

- Output analysis examines the data produced in the simulation and creates a bar chart of the data.
- The bar chart on the side is the chart of the time spent by the domestic passengers in the system.



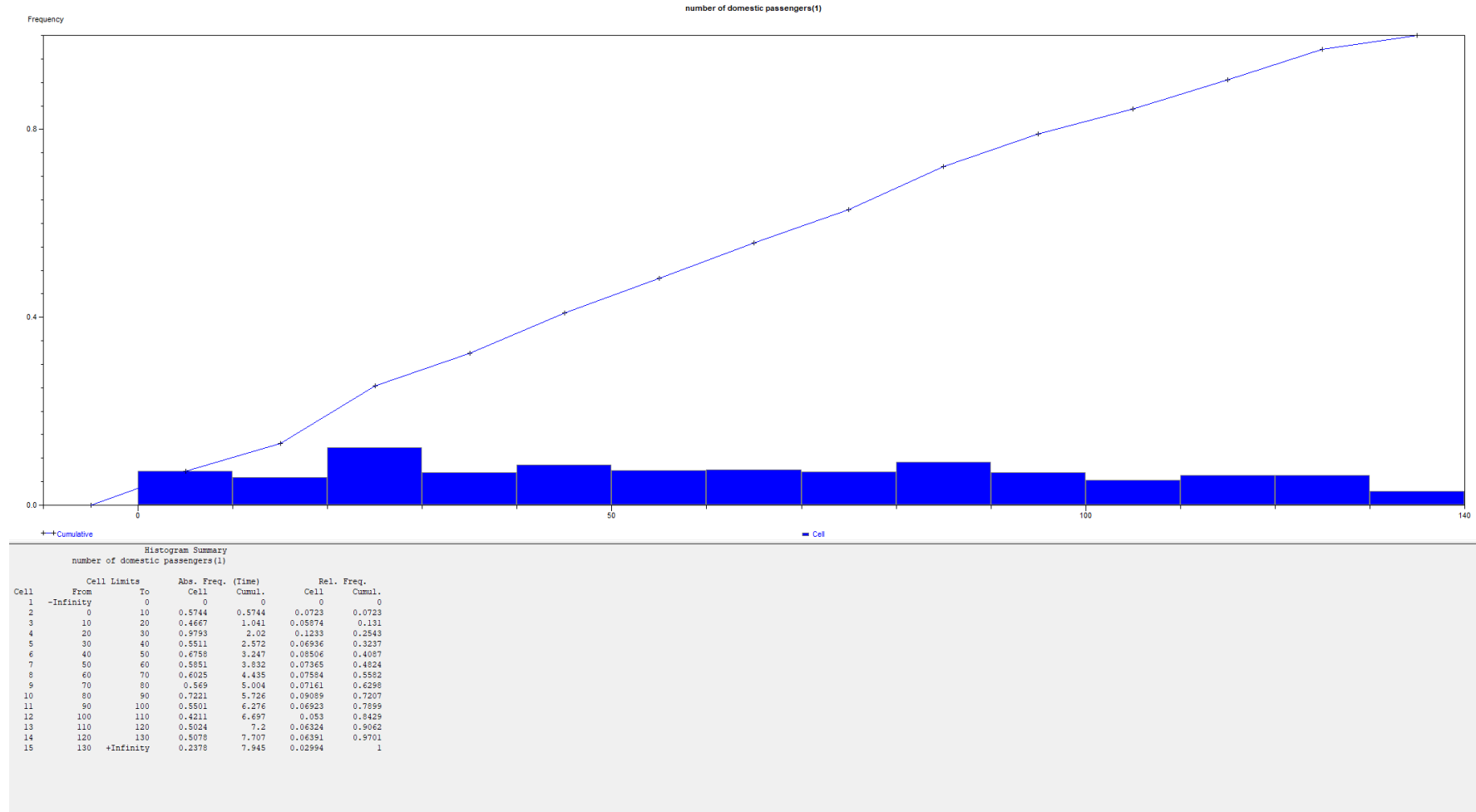
OUTPUT ANALYSIS

- The bar chart on the side is the chart of the time spent by the international passengers in the system.



OUTPUT ANALYSIS

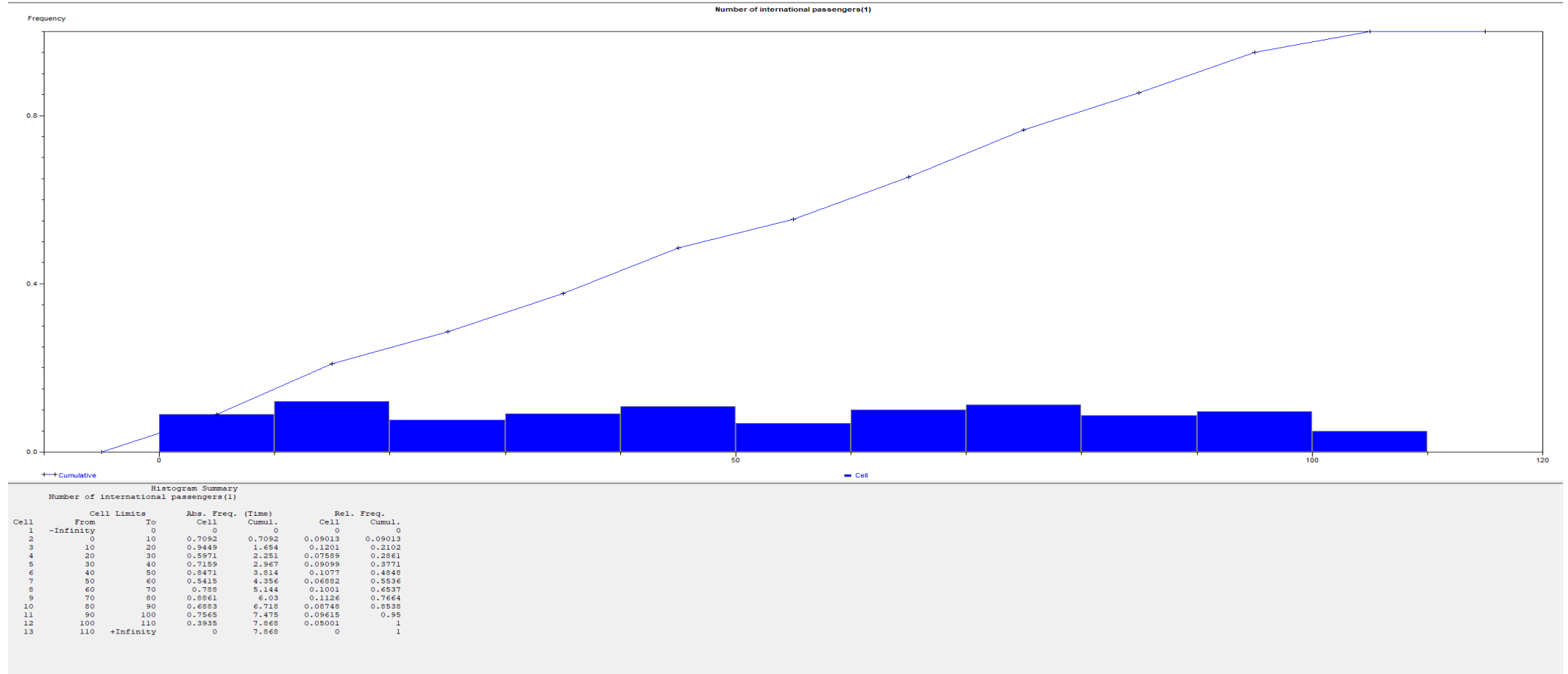
- A histogram was created with the numbers of domestic passengers.



Number of Domestic Passengers Histogram



OUTPUT ANALYSIS

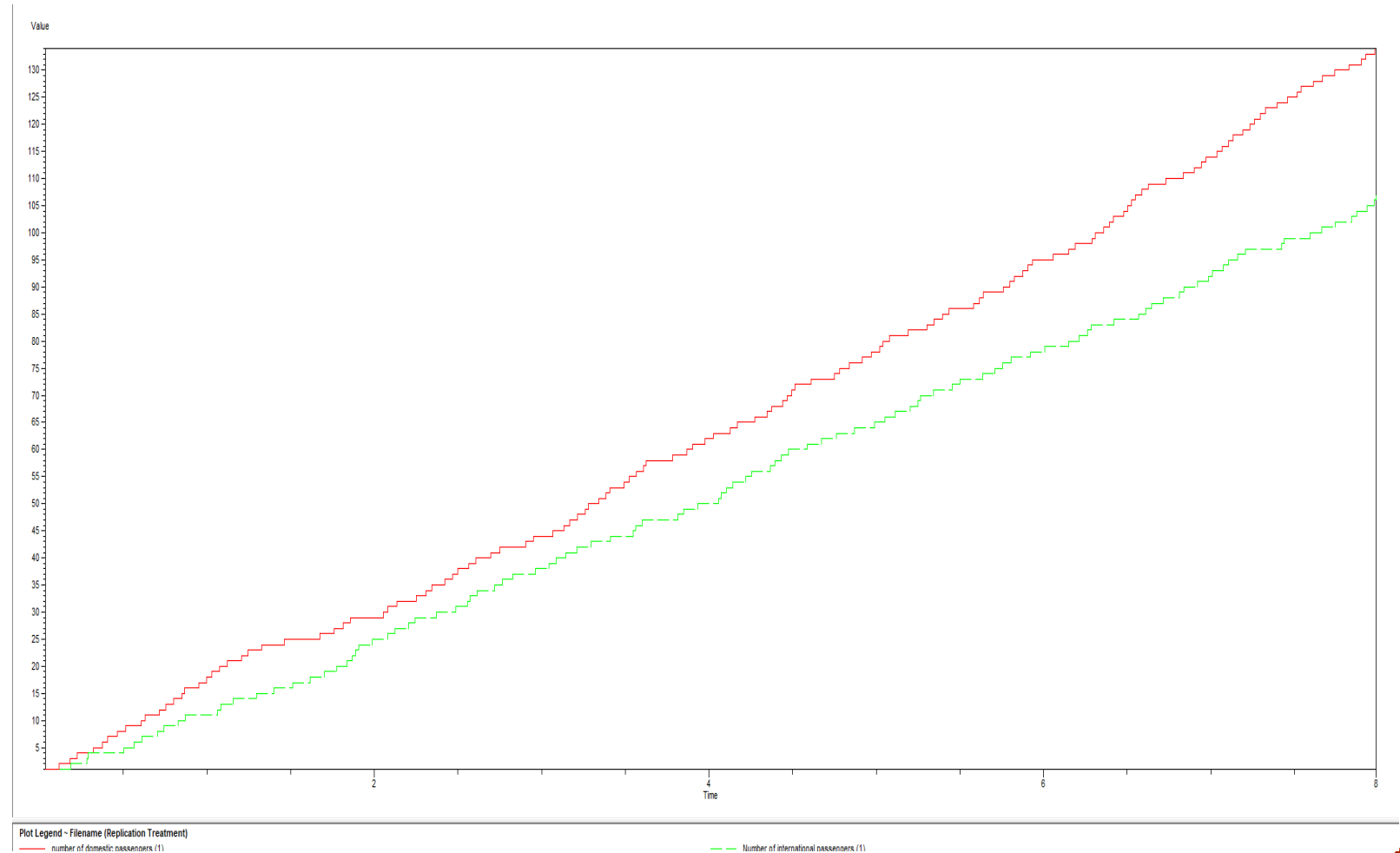


Number of International Passengers Histogram



OUTPUT ANALYSIS

- Comparison of number of international and domestic passengers



Thank you
for
listening!

